



OctoLearn Intelligence Report

Generated By	OctoLearn AutoML	Date	February 22, 2026 at 15:35
Task Type	Classification	Dataset Size	500 rows x 30 features
Risk Score	20/100 (Low Risk)	Best Model	Logistic Regression

This report was auto-generated by OctoLearn's intelligent pipeline. It narrates the complete journey of your data -- from raw input through cleaning, profiling, model training, and actionable recommendations.



ACTIONABLE RECOMMENDATIONS

Based on the comprehensive analysis, OctoLearn has generated the following prioritized recommendations to further improve your data and models:

ACTION PLAN FOR PRODUCTION

Strategic Business Insights

- Key Lever: mean concave points — Significant statistical weight. Prioritize operational focus on this variable.
- Key Lever: worst concave points — Significant statistical weight. Prioritize operational focus on this variable.
- Key Lever: mean radius — Significant statistical weight. Prioritize operational focus on this variable.

MEDIUM PRIORITY

[MEDIUM] Dataset contains only numeric features. Consider polynomial features or interaction terms to capture non-linear patterns.

[MEDIUM] With 21 numeric features, consider: polynomial features, interactions between important features, and feature scaling/normalization for distance-based models.

INFORMATIONAL PRIORITY

[INFO] No missing values detected in original data. Data quality is excellent.

[INFO] Classification detected. Start with: Logistic Regression (baseline), Random Forest (robust), or XGBoost (best performance). Use F1-score for imbalanced data, accuracy for balanced.

THE DATA STORY

Your dataset arrived with 500 records and 30 features, setting up a classification challenge. OctoLearn's 6-phase intelligent pipeline examined every corner of this data to extract maximum insight.

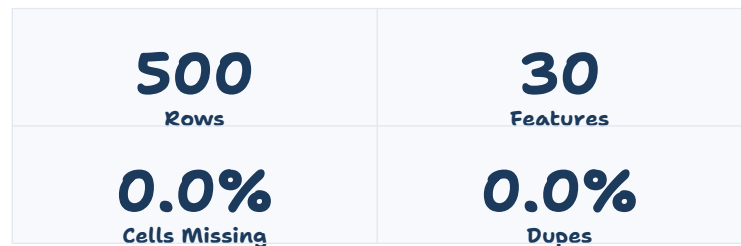
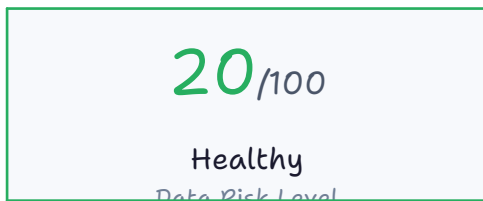
The features break down into 30 numeric and 0 categorical columns. Of the 30 total features, 0 contained missing values that required intelligent imputation strategies.



After deep analysis, OctoLearn assigned a Data Quality Risk Score of 20/100 (Low Risk). This places your dataset in the green zone -- minimal risk factors were detected. Your data is well-structured and ready for modeling.

OctoLearn trained and evaluated 7 models with hyperparameter optimization. After rigorous benchmarking, logistic_regression emerged as the champion with a score of 0.9900. See the Model Arena section for full comparison.

DATA HEALTH DASHBOARD



Data Quality Score: 80.0/100



Risk Factors Identified

Factor	Description	Severity
Imbalance	Severe imbalance ratio: 0.592	High
Low Variance	Low variance columns: 2	Med

DATA TRANSFORMATION JOURNEY

OctoLearn's intelligent cleaning pipeline automatically transforms your raw data into a clean, model-ready dataset. Here is the before-and-after comparison:

Metric	Before (Raw)	After (Clean)	Change
Rows	500	500	+0
Features	30	21	-9



Numeric Columns	30	21	-9
Categorical Columns	0	0	+0
Cols with Missing Values	0	0	+0

What OctoLearn Did

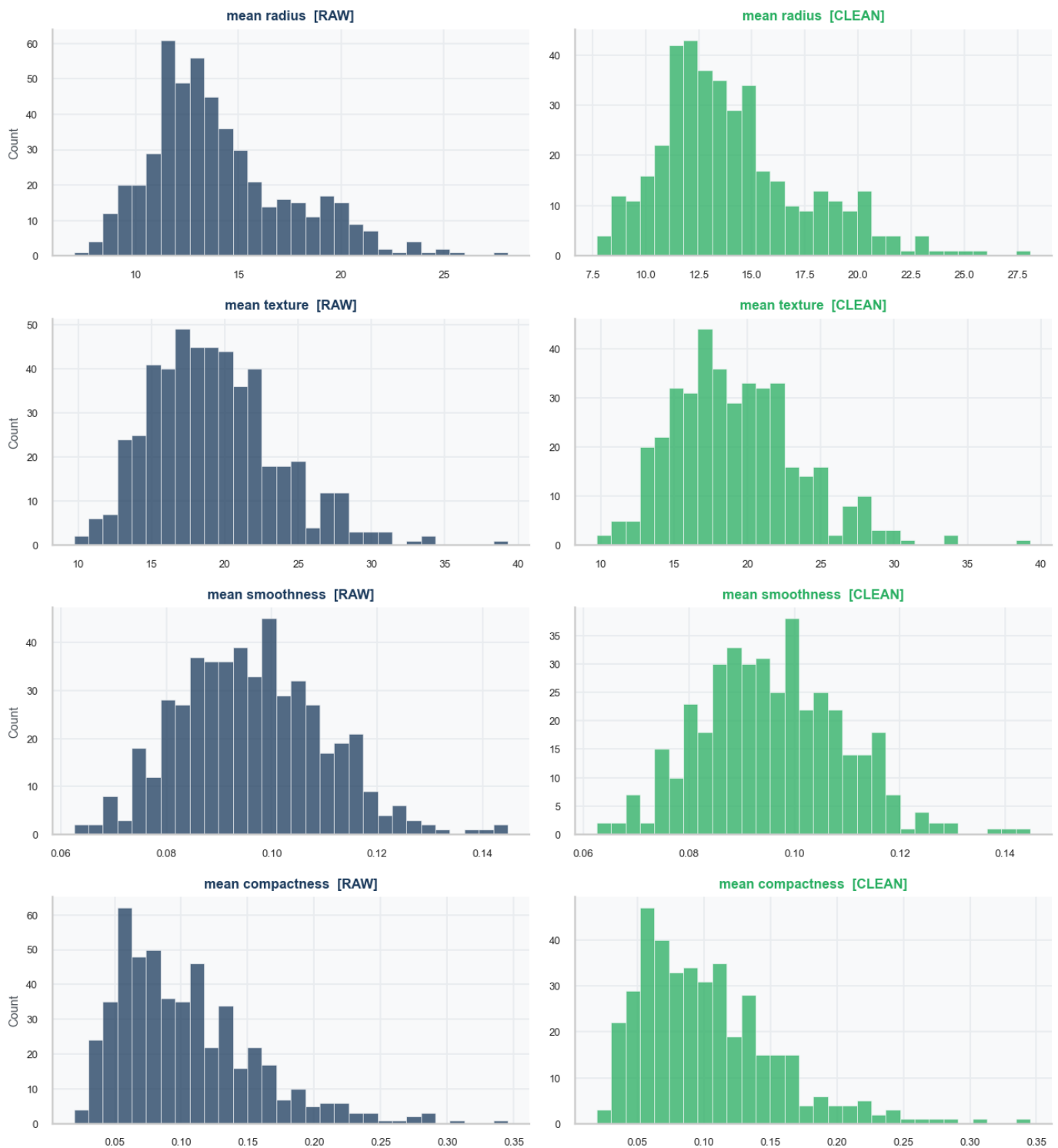
- ☐ Dropped 2 low-variance features
- ☐ Standardized features using standard scaling

Distribution Comparison: Raw vs Cleaned

The histograms below show how OctoLearn's cleaning pipeline changed the distribution of your top numeric features. Navy = raw data, Green = cleaned data.



Feature Distributions: Before vs After Cleaning



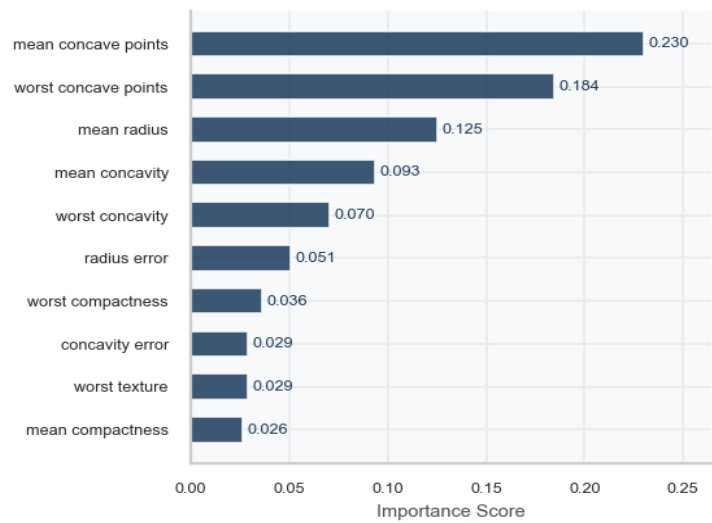
FEATURE INTELLIGENCE

OctoLearn calculated feature importance scores to identify which variables have the strongest predictive power. Higher scores indicate greater influence on the model's



decisions.

Rank	Feature	Score
1	mean concave points	0.2299
2	worst concave points	0.1845
3	mean radius	0.1251
4	mean concavity	0.0935
5	worst concavity	0.0705
6	radius error	0.0510
7	worst compactness	0.0364
8	concavity error	0.0293
9	worst texture	0.0289
10	mean compactness	0.0262

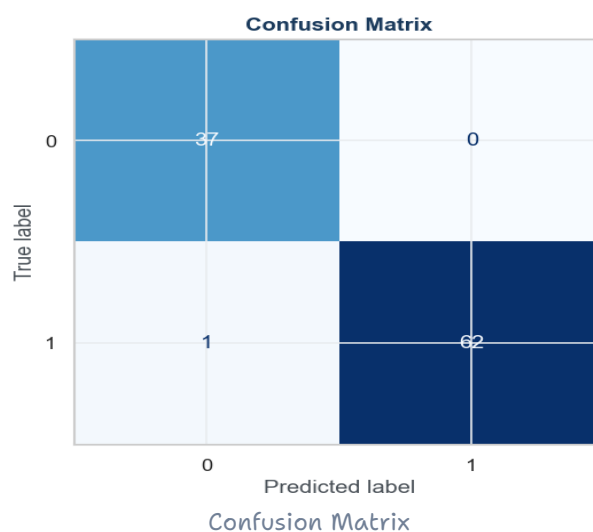


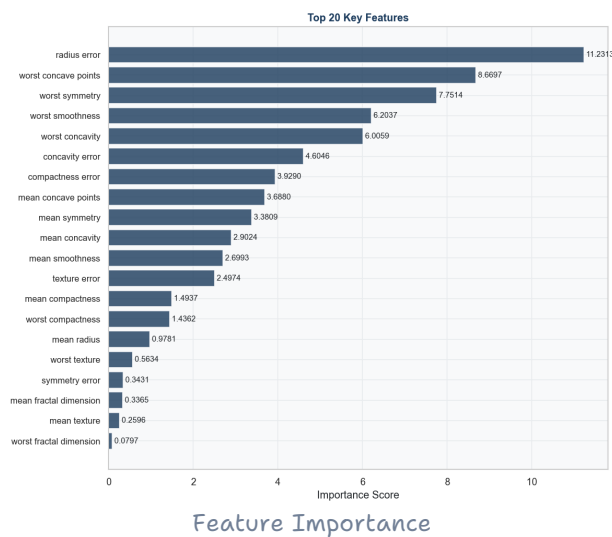
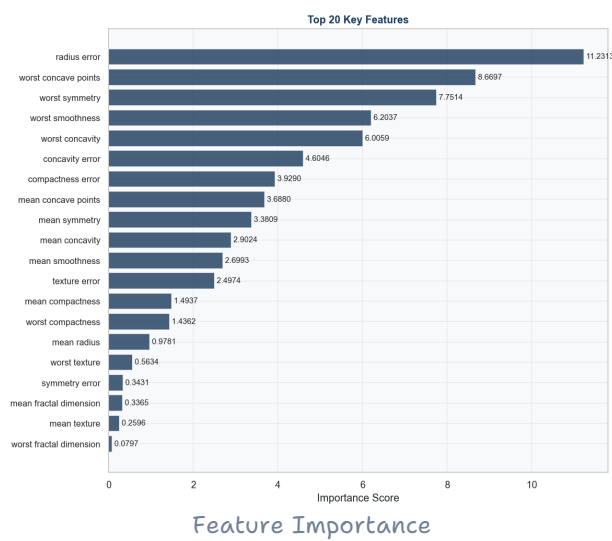
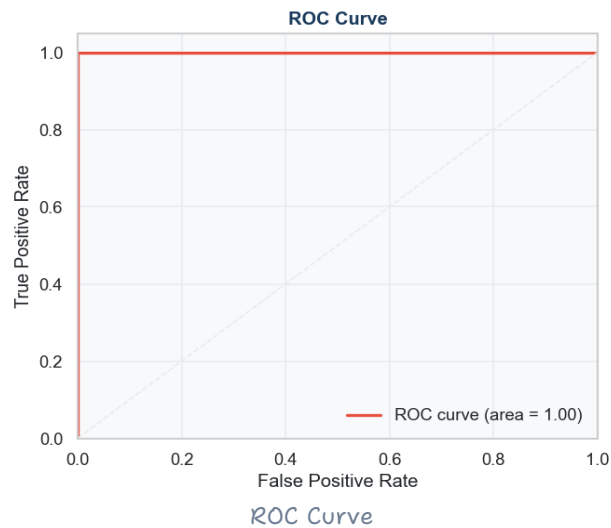
Feature importance scores (higher = more predictive)

MODEL ARENA

Best Model Deep Dive

Detailed performance analysis of the champion model.





OctoLearn trained 7 models with automated hyperparameter tuning. Each model was evaluated on a held-out test set to ensure unbiased comparison. The champion is Logistic Regression with a primary score of 0.9900.



CHAMPION

Logistic Regression

#	Model	Score	Accuracy	Precision	F1	Recall
1	logistic_regression	0.9900	0.9900	0.9903	0.9900	0.9900
2	random_forest	0.9900	0.9900	0.9903	0.9900	0.9900
3	stacking_ensemble	0.9900	0.9900	0.9903	0.9900	0.9900
4	gradient_boosting	0.9801	0.9800	0.9810	0.9801	0.9800
5	lightgbm	0.9801	0.9800	0.9810	0.9801	0.9800
6	svm	0.9800	0.9800	0.9800	0.9800	0.9800
7	xgboost	0.9701	0.9700	0.9704	0.9701	0.9700

Models are ranked by primary optimization score. The #1 model (highlighted) was selected as the best performer.

ADVANCED ANALYSIS

Outlier Detection

Detection Method	Outliers Found
Total Outlier Features	20
Total Outlier Rows	306
Outlier Percentage	61.2
Severity	high



Features With Outliers

```
['mean radius', 'mean texture',  
 'mean smoothness', 'mean  
compactness', 'mean concavity',  
 'mean concave points', 'mean  
symmetry', 'mean fractal  
dimension', 'radius error', 'texture  
error', 'compactness error',  
 'concavity error', 'concave points  
error', 'symmetry error', 'worst  
texture', 'worst smoothness',  
 'worst compactness', 'worst  
concavity', 'worst symmetry',  
 'worst fractal dimension']
```

VISUAL INSIGHTS

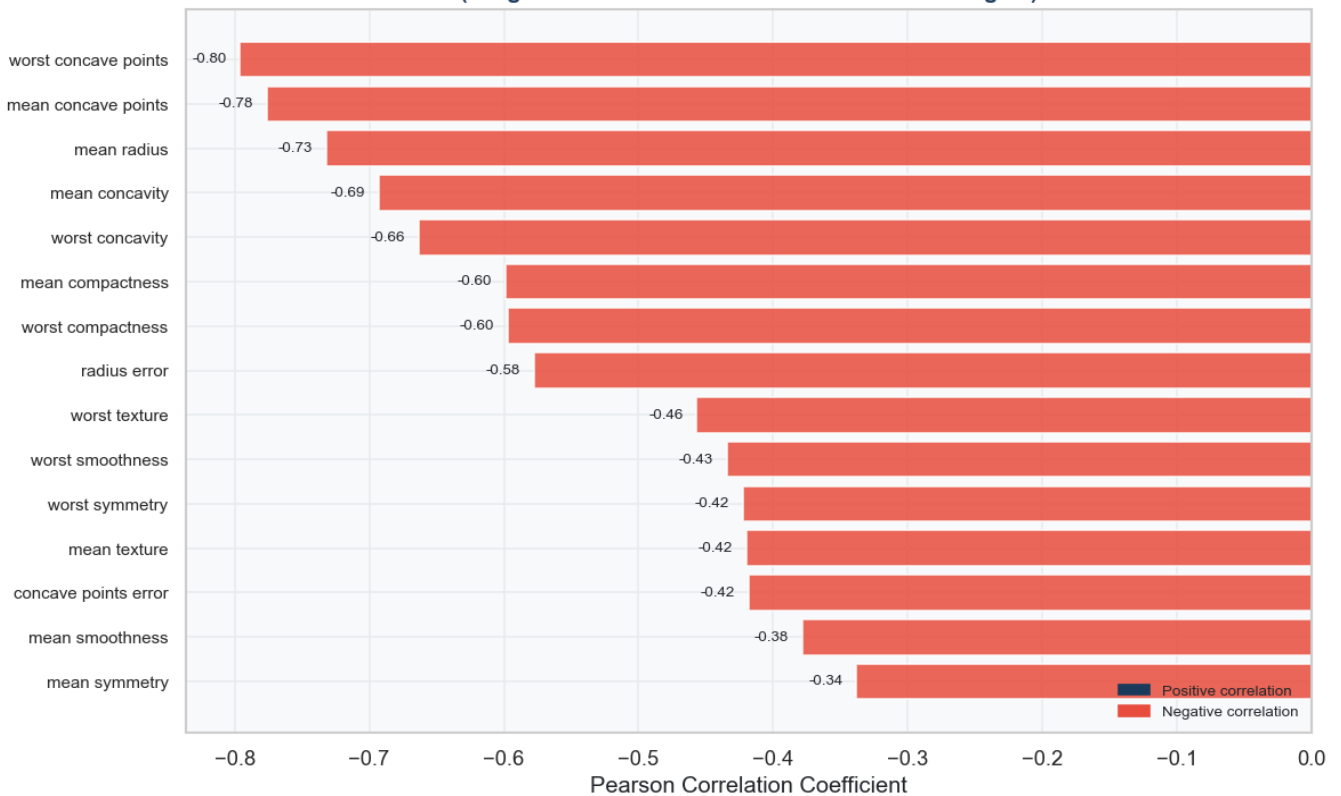
The following visualizations provide deeper insight into feature distributions and correlations within your cleaned dataset.

Goal Correlation Analysis

Analysis shows that worst concave points (strong negatively influence, $r=-0.80$), mean concave points (strong negatively influence, $r=-0.78$), mean radius (strong negatively influence, $r=-0.73$) are the primary predictors for target. Changes in these features are most likely to impact your results. The chart below identifies which of your 21 numeric features most strongly correlate with target. This reveals the key drivers behind your data's goal.



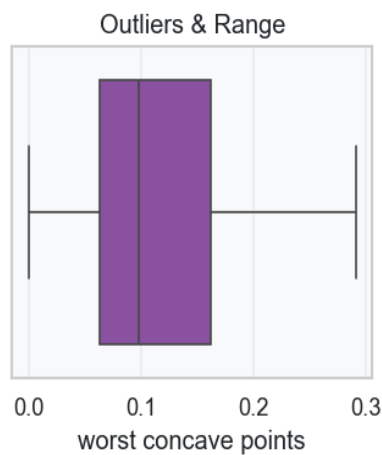
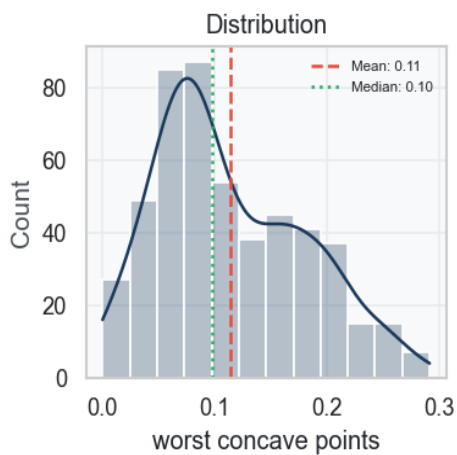
Top 15 Features Correlated with target
(Insights into which features most influence the goal)

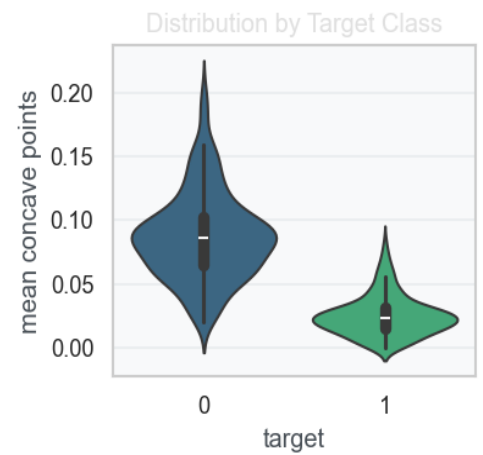
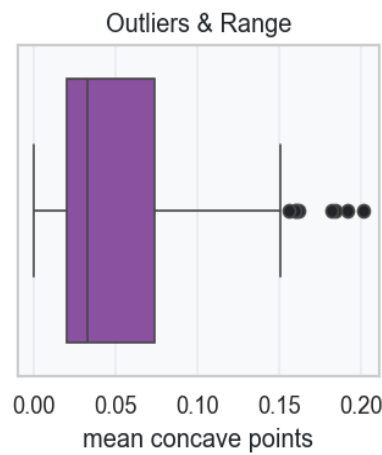
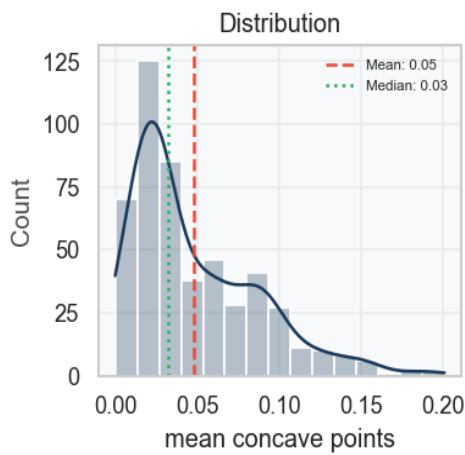
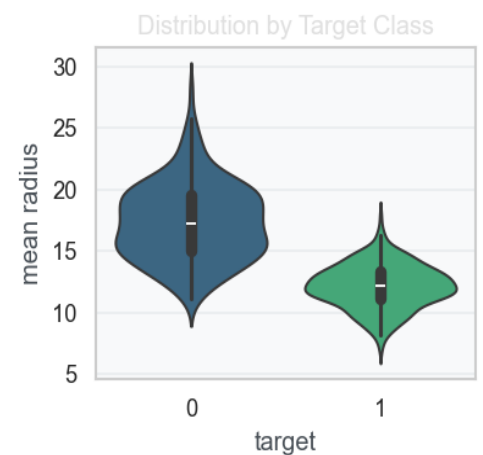
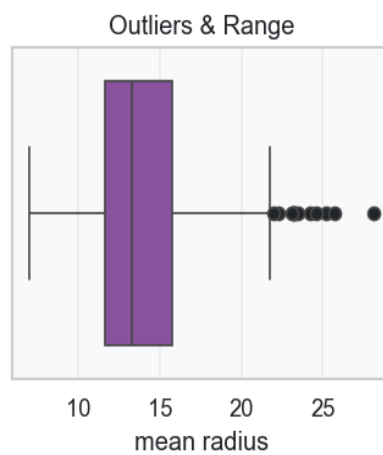
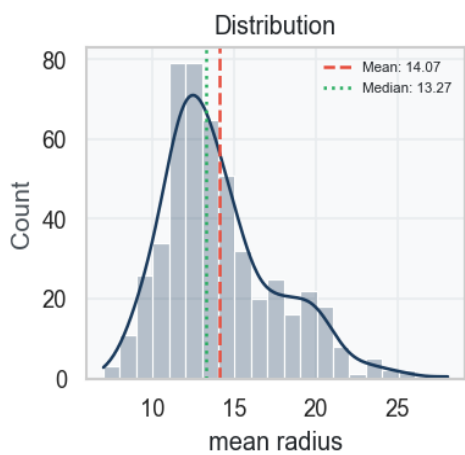
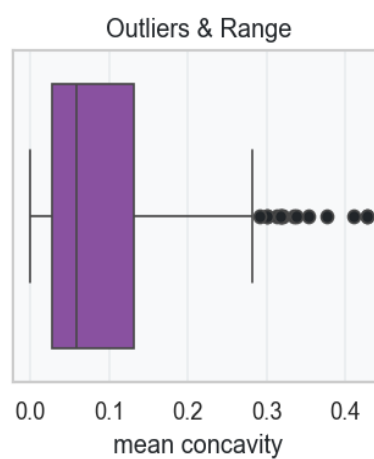
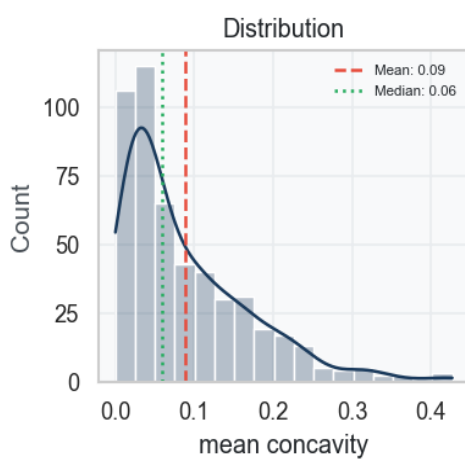


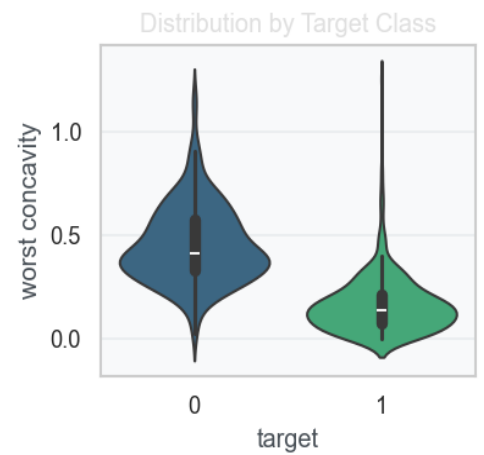
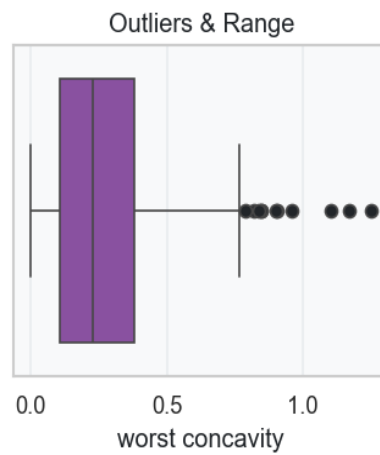
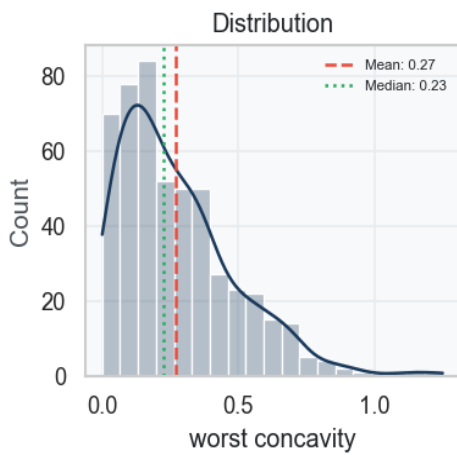
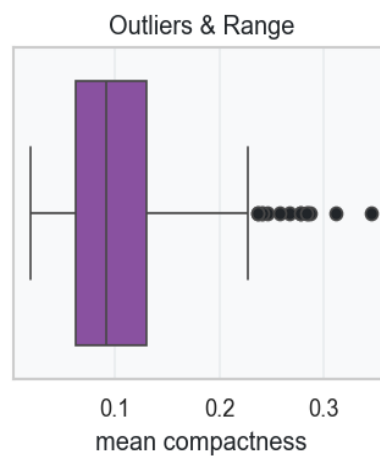
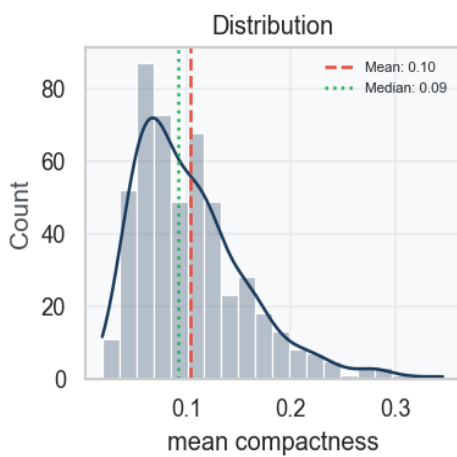
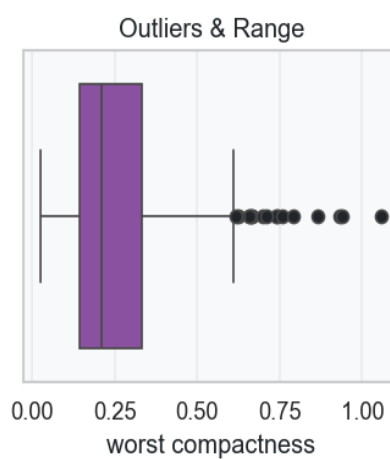
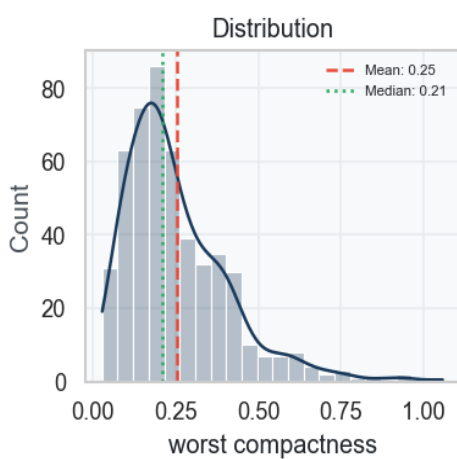
Ranked correlation chart: blue bars = positive correlation, red bars = negative correlation. Higher absolute values indicate features that have more linear influence on the target goal.

Feature Distributions

Feature Analysis: worst concave points



**Feature Analysis: mean concave points****Feature Analysis: mean radius****Feature Analysis: mean concavity**

**Feature Analysis: worst concavity****Feature Analysis: mean compactness****Feature Analysis: worst compactness**



Feature Analysis: radius error

